

Matrix Chain Multiplication Optimization Using Interval Dynamic Programming

Team 09

Project Description

Matrix Chain Multiplication Optimization using Interval Dynamic Programming is a Java-based web application designed to find the most efficient way to multiply a sequence of matrices. Matrix Chain Multiplication does not change the order of matrices, but different parenthesizations can result in significantly different numbers of scalar multiplications.

The project implements **Interval Dynamic Programming** to calculate the minimum multiplication cost for a given matrix chain. It also determines the **optimal parenthesization** and compares the optimized cost with a **naive left-to-right multiplication strategy**.

Key Features

- Accepts the number of matrices and their dimensions as input.
- Dynamically generates matrix representations based on the number of matrices.
- Calculates the minimum number of scalar multiplications using Interval Dynamic Programming.
- Generates the optimal matrix multiplication order.
- Calculates the cost of naive left-to-right multiplication.
- Displays operations saved and efficiency improvement.
- Generates and displays the Dynamic Programming cost table.
- Provides an interactive **3D visualization** of the matrix chain.
- Includes Play, Pause, Previous, Next, and Reset controls for visualization.
- Provides a simple web-based interface for understanding the Matrix Chain Multiplication optimization process.

Technologies Used

- **Java**
- **Spring Boot**
- **Maven**
- **HTML**
- **CSS**

- **JavaScript**
- **Interval Dynamic Programming**
- **REST API**

Algorithm

The project uses the Matrix Chain Multiplication recurrence:

$$\begin{aligned} dp[i][j] = & \\ \min(dp[i][k] + dp[k+1][j] & \\ + p[i-1] \times p[k] \times p[j]) & \end{aligned}$$

The algorithm has:

Time Complexity: $O(n^3)$

Space Complexity: $O(n^2)$

Expected Outcome

The application demonstrates how Dynamic Programming can reduce unnecessary matrix multiplication operations by finding an efficient parenthesization of the matrix chain. It provides both numerical results and visual representation to make the optimization process easier to understand.